TAREK SHEREEN DARWISH

Software Engineer: Systems, Rust & Open Source

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SUMMARY

Systems-minded software engineer finishing a B.Sc. in Intelligent Systems Engineering. Designs and builds low-level Rust and C projects: a Windows game recorder with a GPU-to-disk capture path and an allocation-free DVR buffer, and ESP32 firmware for a health band. Merged fix in librespot (#1786) and three merged PRs to Sonora (Spotify Connect, live Liked Songs, Jams). Winner of the Youth Leaders Foundation (YLF) Hackathon 2026 and 2nd place at Banha, both with Healthica. Looking for a junior or graduate software engineering role, available from February 2027.

TECHNICAL SKILLS

Languages: Rust, C, C++, Python, SQL; also TypeScript/JavaScript, Dart, Kotlin, Java

Systems: Direct3D 11, Media Foundation, WASAPI, Win32 API, BLE (GATT), ESP32 firmware (C/Arduino), llama.cpp

Linux & tools: Linux (Arch), bash and PowerShell scripting, WSL2, Git, Docker, CI/CD, OCI CLI

Cloud & ML: Azure, AWS, Cloudflare, Supabase, TensorFlow Lite, PyTorch, scikit-learn

OPEN SOURCE

- librespot (Rust, Spotify client library): merged PR #1786, github.com/librespot-org/librespot/pull/1786. Liked Songs updates arrive as plain JSON but were decoded as base64 and dropped; the fix passes them on as JSON. Small change (+6 -3 lines).
- Sonora (Rust music app), 3 merged PRs (Oct 2026): Spotify Connect device #895 (+2,809 -93, 51 files), live Liked Songs #905, Spotify Jams #934. Open: right-to-left text #880 with sonorahq/gpui #18.
- Also open: librespot #1804 (docs for Spotify Jam endpoints), MeasureIt_ARCH #309 (Python, Blender add-on).

PROJECTS

ZeroClip: Windows Game Clipper · *Creator and sole developer · pre-alpha · Rust, Direct3D 11, Media Foundation, WASAPI*

- GPU screen capture feeding hardware H.264/HEVC encoders with frames kept in VRAM, WASAPI audio, an allocation-free, GOP-aware DVR buffer, and a fragmented MP4 writer that saves clips without re-encoding.

Healthica: Smart Band Health Monitor · *Winner, YLF Hackathon 2026 (team of 5) · 2nd place, Banha Hackathon · C, ESP32, BLE, TFLite*

- Worked on the ESP32 firmware: reads a MAX30105 PPG sensor and a GSR input, detects beats, computes HRV (RMSSD, SDNN, pNN50) and sends a payload of about 250 bytes over BLE GATT notifications, with a WiFi HTTP fallback.
- Split the work to keep the band light: HRV and PPG features on the ESP32, inference on the phone with INT8 TensorFlow Lite glucose models of 8 to 23 KB. Also built the Flutter app and web dashboard.

safi: Local Agentic LLM Backend · *Python, llama.cpp, PowerShell · in progress since Oct 2026*

- Planner and executor model pair (Qwen3.8-27B, Qwen3.6-35B-A3B) on llama.cpp, 16 GB VRAM laptop. Built an onboarding probe that checks every layer is on the GPU and measures VRAM, prefill and decode speed, plus a 10 minute thermal soak.
- Also built a tool registry, a CPU embedder and PowerShell/shell router supervisors, with a pytest suite.

OpenCrosshair: Windows Crosshair Overlay · *Open source (MIT) · github.com/Tu2525/OpenCrosshair · Rust, Win32 API, egui*

- Click-through overlay with per-game presets. No DLL injection by design; tradeoff: no exclusive fullscreen.

EXPERIENCE

Generative AI Engineer, Digital Egypt Pioneers Initiative — *Cairo, Egypt (Hybrid) · Nov 2025 – Jul 2026*

- Built and deployed (Docker, Hugging Face Spaces) an explainable fraud-detection service: a 310-feature classifier behind FastAPI with Gemini-written explanations from SHAP.

Backend & AI Engineer Intern, Aispeak — *Madrid, Spain (Remote) · Jul 2025 – Jan 2026*

- Built a backend pipeline routing user messages to a LLaMA-70B model on Azure, plus Supabase real-time messaging.

EDUCATION & CERTIFICATIONS

B.Sc., Intelligent Systems Engineering, Helwan National University — *Helwan, Egypt · 2022 – Jan 2027 (expected)*

DEPI Generative AI and Mobile Tracks | AWS Cloud Practitioner | Huawei HCIP-AI | Huawei ICT Associate | Stanford ML Specialization